

Measuring Cognitive Load in Introductory CS: Adaptation of an Instrument

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ABSTRACT

A student's capacity to learn a concept is directly related to how much cognitive load is used to comprehend the material. The central problem identified by Cognitive Load Theory is that learning is impaired when the total amount of processing requirements exceeds the limited capacity of working memory. Instruction can impose three different types of cognitive load on a student's working memory: intrinsic load, extraneous load, and germane load. Since working memory is a fixed size, instructional material should be designed to minimize the extraneous and intrinsic loads in order to increase the amount of memory available for the germane load. This will improve learning. To effectively design instruction to minimize cognitive load we must be able to measure the specific load components for any pedagogical intervention. This paper reports on a study that adapts a previously developed instrument to measure cognitive load. We report on the adaptation of the instrument to a new discipline, introductory computer science, and the results of measuring the cognitive load factors of specific lectures. We discuss the implications for the ability to measure specific cognitive load components and use of the tool in future studies.

Categories and Subject Descriptors

K.3.2 [Computers and Education]: Computer and Information Science Education—*computer science education*

Keywords

cognitive load theory, measuring cognitive load, confirmatory factor analysis, survey

1. INTRODUCTION

Cognitive Load can be defined as “the load imposed on an individual's working memory by a particular (learning) task” [42, p.599]. The resultant performance for a student to learn a specific concept is directly related to how much cognitive load is used to comprehend the material. If instruction

overloads the student's working memory, then knowledge retention and any possible transfer will suffer. As designers of instructional material, it is our responsibility to ensure that we do not overload the learner's working memory where possible when presenting new material. That is, we should help ensure that students' attentional abilities are directed to key aspects of the content and learning activities that have maximum value, rather on than extraneous aspects of the material. But how can we know if we are successful?

The central problem identified by Cognitive Load Theory (CLT) is that learning is impaired when the total amount of processing requirements exceeds the limited capacity of working memory [30]. According to Cognitive Load Theory [34, 36, 44], instruction can impose three different types of cognitive load on a student's working memory: intrinsic load, extraneous load, and germane load. Intrinsic load (IL) is defined as a combination of the innate difficulty of the material being learned as well as the learner's characteristics [20]. A topic is considered to have a high IL if the material being learned is interconnected; that is, learning requires processing several elements simultaneously to understand their relations and interactions [35]. Intrinsic load can also vary with the domain expertise and previous knowledge of the learner [37] in that learners with a higher level of previous knowledge may chunk the material differently than novices [6], allowing them to hold more information in working memory. Extraneous load (EL) is the load placed on working memory that does not contribute directly toward the learning of the material—for example, the attentional resources consumed while understanding poorly written text or diagrams without sufficient clarity [20]. The IL and EL are the factors that can be controlled through instructional design. The final category is that of germane load (GL) which are the instructional features that are *necessary* for learning the material [20].

One of the assumptions of CLT is that these three components are additive [30]; thus if the extraneous load is using the capacity of working memory, little can be devoted to the germane load. Because working memory is considered to be a fixed size [22, 5], it behooves the instructional designer to minimize the EL, design appropriately for the IL, and emphasize the GL. To accomplish this, we must be able to measure the specific load components for any pedagogical intervention. Until recently, there were no effective instruments to measure the different components of cognitive load.

This paper reviews previous attempts to measure cognitive load as both an overall entity and with respect to its defined factors. It then describes the specifics of the study conducted. The statistical results are reported and the pa-

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per concludes with a discussion on the implications of being able to measure cognitive load components within introductory computer science. We also describe future use for the instrument in measuring pedagogical interventions as well as limitations of the survey.

2. MEASURING COGNITIVE LOAD

Since the discovery and identification of CLT, researchers have searched for a means to measure cognitive load. To date, this has been accomplished through indirect, subjective, and direct measures.

2.1 Indirect Measures of CLT

Researchers began exploring cognitive load when they saw how problem-solving activities actually interfered with learning. Computational models provided independent evidence that working memory was strained to accomplish both problem-solving and learning, so decreasing cognitive load became an important instructional design goal. Problem solving that required more searching of knowledge led to inferior learning outcomes [37]. This led Sweller and his colleagues [33, 3] to develop production system models to simulate the problem solving using both high-intensive and low-intensive search strategies. Results demonstrated that high-intensive search methods required a more complex model to simulate the problem solving process.

Another indirect measure of CLT is based on learner performance indicators during the acquisition of the material. Without a direct measure, Chandler and Sweller [9, 10] used instructional time as a proxy for cognitive load. Error rates were higher during acquisition of knowledge under conditions in which the expected time to solve was higher. This corresponded to their measurement of a higher cognitive load. Error rates have also been used to identify differences in cognitive load within problems [3, 4].

2.2 Subjective Measures of CLT

In 1992, Paas theorized that learners are able to assess the amount of mental effort required during learning and testing and that this ‘intensity of effort’ may be considered to be an ‘index’ of cognitive load [26, p.429]. A 9-point Likert scale ranging from very, very low mental effort (1) to very, very high mental effort (9) was used to ask learners to rate their mental effort at various points during the learning and testing cycle. Paas found that there was a correlation between self-rated mental effort and test performance. A follow-up study [28] replicated the findings and also found that the subjective ratings were more sensitive and less intrusive than an objective physiological measure also captured during this study. The 9-point scale was also found to be highly reliable [29]. The success of these initial instruments to measure cognitive load led others to adopt the subjective scale. However, during the adoption process, the wording of survey questions were changed and the term *mental effort* was changed to *difficulty* or *easy*, thus measuring something different. While subjective measures of difficulty and mental effort may be related, they are not interchangeable; difficulty does not always match effort [41].

However the subjective rating scale, regardless of the wording used, has been shown to be the most sensitive measure available to differentiate the cognitive load imposed by different instructional methods [37]. The subjective measures have also been consistent in matching performance data predicted by CLT [23]. The subjective scale has been used extensively to measure the relative cognitive load of

different instructional methods with over 25 studies having used it between 1992 and 2002 [25].

Building on the initial self-rating scale, Paas and van Merriënboer developed an efficiency measure for cognitive load [27]. This efficiency measure combined both mental effort with task performance indicators. This measurement allowed instructors to determine whether specific pedagogical interventions yielded high or low instructional efficiency results. Over 30 cognitive load theory related studies have used this efficiency measure [41]. However the adoption of this measurement tool also resulted in changes from the original use: measuring the mental effort changed the efficiency of learning. Obviously measurement in and of itself adds to cognitive load for participants. Also, *when* the mental effort was measured changed what the efficiency was measuring. For those studies where mental effort was measured immediately following the acquisition phase and prior to the testing phase, training efficiency was measured. For those studies where mental effort was measured after test performance, learning efficiency was measured.

2.3 Direct Measures of CLT

Two basic means of measuring cognitive load through direct measures have been used: using a dual task and physiological measurements. A secondary or dual task study requires learners to engage in an additional cognitive activity that is secondary to the primary task of learning. If a higher cognitive load is required for the primary task, performance on the secondary task will suffer. Usually the secondary task is quite dissimilar and requires less working memory than the primary task, such as recognizing when a letter changes color or when a specific tone is heard. Examples of cognitive load studies using the dual task methodology can be found in [8, 11, 40]. In general, CLT research has made far less use of the dual task method than the subjective method for measuring cognitive load. However the advantage of using a dual task method is that it can provide an almost continuous measure of cognitive load during a task.

Physiological measures can also provide a continuous measure of cognitive load during a task. Researchers have used measurements of heart rates [28], pupillary response [39], EEGs [1], and eye tracking [43, 38]. Others have even advocated using fMRI [45]. In general the results support measuring CLT using physiological measures, but the studies have only been run in laboratory settings due to the requirement of specialized equipment. This raises questions about the ecological validity of such studies and highlights the impracticality of using these approaches in most applied educational research.

2.4 Critique of CLT Measurements

The basic question of a cognitive load measurement method is whether it is valid, reliable, and practical. Subjective rating scales are simple and practical to use and have proven reliable through repeated use. However, they only deliver a one point post hoc assessment of the cognitive load imposed by the learning task. It remains unclear which of the specific aspects of the learning situation caused the level of cognitive load reported by the student. Although it is assumed that learners are able to be introspective about their own cognitive processes and quantify their perceived mental load during learning, the measures are unable to provide information regarding which processes caused the perceived amount of mental load. It is also not possible to determine

which of the three types of load (IL, EL, or GL) originated the report of mental effort.

Objective methods yield data for more than a single point in time by providing information about instantaneous, peak, average, accumulated, and overall load [1]. However these measurement techniques are far from practical for the majority of any large scale study based on the specialized equipment needed. Objective measurement methods also suffer from the ability to determine which type of load was responsible for the resultant physiological changes.

Several researchers have attempted to distinguish between and measure the different types of cognitive load. Ayres attempted to keep the extraneous cognitive load (EL) constant between treatments thus attributing the differences to a change in intrinsic load (IL) [2]. DeLeeuw and Mayer used a mixed approach (both subjective measures and a secondary task method) to investigate if different instruments could measure the three loads separately [14]. The results indicated that different measures do tap into different processes and show varying sensitivities.

A widely used multidimensional scale is the *NASA Task Load Index (NASA-TLX)* [17], which consists of six subscales that measure different factors associated with completing a task. An overall measure of task load is achieved by combining the subscales of the NASA-TLX. One of those subscales is mental demand/load. However the NASA-TLX was designed for use with interface designs and specifically for the aeronautical industry. In an attempt to measure the different cognitive load categories, Gerjets, Scheiter, and Catrambone selected three items from the NASA-TLX associated with task demands [15, 16]. The researchers argued that the three items selected (mental and physical activity required, effort to understand the contents, and navigational demands of the learning environment) could be mapped to the intrinsic, germane, and extraneous loads, respectively. The test manipulated the complexity of worked examples. There was broad agreement with the test performance data in that groups with the highest learning outcomes reported the lowest cognitive load. However there was no corroborating evidence that the three measures corresponded to the different types of cognitive load as proposed.

However in 2013, Leppink et al. [20] developed an instrument specifically for measuring different types of cognitive load which consists of a ten question subjective survey. (We will refer to this survey as the *Cognitive Load Component Survey*.) The researchers developed the questions and tested them using a set of four studies. The overall purpose of the studies was to compare their instrument as a measurement tool for the three types of cognitive load to other existing subjective measurement tools. The first study used exploratory analysis to determine if the questions developed did indeed load onto the three types of cognitive load (IL, EL, and GL). The second study used confirmatory factor analysis (CFA) to test the existing measurement tools [2, 12, 32, 26] for measurement of specific cognitive load factors. This study revealed that none of the existing survey tools adequately separated the three types of cognitive load in that each had significant cross-loading between factors. The newly developed Cognitive Load Component Survey was also tested using confirmatory factor analysis in the third study. The final study then used the Cognitive Load Component Survey to examine the effects of experimental treatment and prior knowledge on the cognitive load components and learning outcomes of students within a statistics

course. The results of the final study were consistent with outcomes based on CLT.

Leppink et al. [21] recently extended their 2013 work by adapting the survey instrument to another domain, that of learning languages, and replicated their analyses. These new findings reinforce the strong support for the survey measuring both intrinsic and extraneous load, but found less support for the direct measure of germane load.

Here, we report on a study to adapt the Cognitive Load Component Survey for use in an introductory computer science context. We detail how the instrument was adapted to a different discipline and the results of measuring the cognitive load factors of specific lectures during the course.

3. METHOD OF STUDY

Development and initial validation work on the original Cognitive Load Component Survey identified a set of three underlying dimensions (factors): 3 items related to intrinsic load, 3 items measuring extraneous load, and 4 items measuring germane load [20, 21]. Participants respond to each item using an 11-point semantic differential scale from 0 to 10 anchored at “0-not at all the case” and “10-completely the case”. Multiple CFA studies using data from different undergraduate statistics lectures confirmed that this three factor model consistently performs well [20, 21].

Given the consistent findings related to using the cognitive load questionnaire in statistics lectures, we sought to adapt and verify its applicability for computer science. Since the underlying cognitive load theory dimension tied to each factor should be independent of any particular discipline, our method for validating this adaptation hinges on verifying that the original factor model holds true for the newly reworded items in the new disciplinary context of computing. That is, the structure of items on the questionnaire should be robust to slight alterations in question wording better suited to terminology used in computer science coursework.

Similar to the changes done by [21], the wording was changed in a total of three questions. In questions 2 and 9 the word “formulas” was changed to “program code”. In question 8, the word “statistics” was changed to “computing / programming”. These word changes were piloted with a small group of students to determine if they were understood by participants and were appropriate for the concepts addressed during lectures. The final modified instrument instructions and items are provided in Figure 1.

As posited by Leppink et. al. [20], items 1, 2, and 3 measure the intrinsic load (IL); items 4, 5, and 6 are the extraneous load (EL) factors; and items 7 through 10 measure the germane load (GL). Note that the wordings for items 1 through 6 are negatively worded, in the sense that a response of “10-completely the case” indicates a very high detriment to learning. This is in comparison to items 7 through 10 which are positively worded. Because items 1 through 6 are measuring factors (IL and EL) that we want to minimize, the higher the response indicates that more working memory is being allocated toward undesirable components; while higher scores on items 7 through 10 indicate a desirable use of working memory.

The CS Cognitive Load Component Survey (CS CLCS) was administered twice during the term of an introductory course in computing using Python designed for non-CS majors that utilizes a media computation context. The students were declared majors in Liberal Arts (mostly literature, public policy, and international affairs), Business, and Architecture. Data was collected from two different sec-

Instructions: All of the following questions refer to the lecture that just finished. Please respond to each of the questions on the following scale by circling the appropriate number (0 meaning not at all the case and 10 meaning completely the case):

1. The topics covered in the activity were very complex.
2. The activity covered program code that I perceived as very complex.
3. The activity covered concepts and definitions that I perceived as very complex.
4. The instructions and/or explanations during the activity were very unclear.
5. The instructions and/or explanations were, in terms of learning, very ineffective.
6. The instructions and/or explanations were full of unclear language.
7. The activity really enhanced my understanding of the topic(s) covered.
8. The activity really enhanced my knowledge and understanding of computing / programming.
9. The activity really enhanced my understanding of the program code covered.
10. The activity really enhanced my understanding of the concepts and definitions.

Figure 1: CS Cognitive Load Component Survey

tions of the course. Both sections were taught by the same instructor and covered the same material on the days of collection. The first dataset (Lecture 1) was collected mid-way through the course. Students had already completed several assignments involving text and list manipulations and image processing. The first dataset was collected after a lecture on generating HTML for web pages using Python functions. The second dataset (Lecture 2) was collected in the last 20% of the course and followed the initial lecture on sound processing. Students saw visualizations of different sounds, heard an explanation for how sound is digitized, and saw demonstrations of code for manipulating volume.

Surveys were paper-based and distributed at the end of each lecture. Approximately 10 minutes at the end of class was allocated to explaining the purpose of the survey, distribution and collection of the instrument. The survey items were presented in three different orders (see Table 1). The three versions were put into randomized order so that people sitting next to each other were not necessarily answering the questions in the same order. This mitigates ordering effects within the questions that might skew the results.

After each collection of data, surveys with obvious patterns (e.g., all 5's, zig-zag responses) were filtered out to ensure participants had appropriately considered each question prompt. For Lecture 1, only one invalid survey was removed from processing, and for Lecture 2, two invalid surveys were eliminated. The survey yielded a total of 156 valid responses from both sections following lecture 1 and a total of 117 valid responses following lecture 2.

Table 1: Question Ordering

order	Item Order	Count in Lecture	
		L_1	L_2
A	1, 7, 4, 2, 8, 5, 3, 9, 6, 10	53	40
B	6, 10, 9, 3, 5, 8, 2, 7, 1, 4	52	43
C	9, 3, 6, 8, 2, 4, 10, 5, 7, 1	51	34

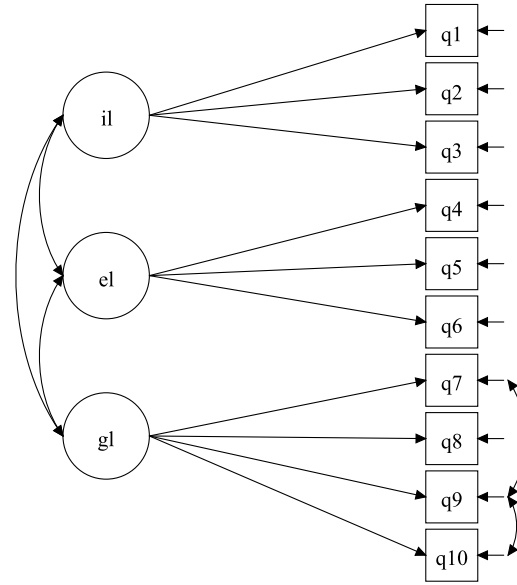


Figure 2: Factor Model

Each data set was then analyzed using confirmatory factor analysis (CFA). CFA allows the researcher to propose and test underlying relationships between items on a survey [7]. It is commonly used within instrument validation to determine whether groups of survey questions that theoretically relate to one another also relate statistically. CFA uses the co-variance matrix of item responses to investigate the degree to which a specified model explains the observed variation in the data set. It produces a set of model fit statistics and parameter estimates that are interpreted to evaluate how well a proposed model captures these relationships.

This data was analyzed using MPlus for confirmatory factor analysis. We replicated the factor model found by the earlier work (see Figure 2) [20]. In this model each item serves as an indicator of a single factor (i.e., it loads on only one factor), and the error terms of two pairs of items to covary (q7 & q9, q9 & q10) due to similarities in the wording of question stems. While Leppink et. al. [20] failed to indicate the statistical estimation algorithm used in their CFA on the original questionnaire, it is probable that maximum likelihood (ML) was used as it is the default in most CFA software packages and is the one most commonly used [7, 19]. Thus, we employed the ML estimator in our analysis. The results of the CFA analysis for the adapted questionnaire are outlined in the following section.

4. RESULTS

4.1 Lecture 1

Fit statistics provided by confirmatory factor analysis on data collected following lecture 1 (N=156) indicated that the model specified in Figure 2 fit the data well ($\chi^2(30) = 36.92, p = 0.18; RMSEA = 0.04, p_{close} = 0.67; CFI = 0.99; TLI = 0.99$). The χ^2 test yielded a non-significant p-value, indicating that the observed co-variance matrix was not significantly different from the expected matrix specified by the model. Further, this model exhibited other fit statistics meeting or exceeding cutoff criteria for well-fitting models specified by Hu and Bentler [18]. The root mean

Table 2: Factor Loadings and Reliability (Lecture 1)

Factor/Item	Factor Loading	Std Err	p-Value
IL - Intrinsic Load ($\alpha = 0.85$)			
Q1	0.75	0.043	< 0.001
Q2	0.93	0.031	< 0.001
Q3	0.76	0.040	< 0.001
EL - Extraneous Load ($\alpha = 0.80$)			
Q4	0.92	0.030	< 0.001
Q5	0.71	0.046	< 0.001
Q6	0.69	0.048	< 0.001
GL - Germane Load ($\alpha = 0.92$)			
Q7	0.91	0.023	< 0.001
Q8	0.86	0.027	< 0.001
Q9	0.87	0.030	< 0.001
Q10	0.80	0.034	< 0.001
Residual Covariance			
Q7 with Q9	-0.19*	0.157	0.230
Q9 with Q10	0.40*	0.091	< 0.001

*denotes a correlation rather than a loading

Table 3: Factor Correlations (Lecture 1), all significant at $p \leq 0.001$

	IL	EL	GL
IL	1.0	0.435	-0.282
EL		1.0	-0.749
GL			1.0

square error of approximation (RMSEA) was less than 0.06, and both the comparative fit index (CFI) and Tucker-Lewis Index (TLI) are greater or equal to 0.95.

Table 2 presents standardized item loadings for each of the factors in the well-fitting model. Loading values range from 0.0–1.0 and indicate the degree to which an item serves as a measure of a factor. Values closer to one signify that the item is a pure measure of that factor. Here we note that all items load significantly and highly (≈ 0.7 or more) on their corresponding factor. Further, Cronbach's alpha values for each factor met or exceeded 0.8, indicating good internal consistency within each factor, and the overall survey internal reliability was $\alpha = 0.89$. The two residual co-variance paths in the model did not exhibit high correlations, as had been the case in some of the original development studies by [20]. Inspection of the modification indices revealed no suggestions that would markedly improve model fit.

Theoretically, factors that measure EL and GL should be negatively correlated. The more EL that exists, there is less working memory available for GL. Components that measure IL and GL should have a correlation around 0 indicating that the relationship between IL and GL is non-linear. Extremely low or extremely high levels of IL may lead to a lower GL score. If a learning task is too easy for a student, the explanations and instructions in the task may not contribute to actual student learning. On the other hand, if the learning task is too complex for a particular student, working memory available for GL activity may be limited.

The observed inter-factor correlations (see Table 3) support this theoretical interpretation. Overall, observed correlations were all less than 0.8, indicating a reasonable degree of discriminant validity (i.e., each factor measures a unique facet of cognitive load) [7, 24]. We observed a weak, positive correlation between intrinsic and extraneous load and a moderate, negative correlation between extraneous and germane load. Negative correlations in the table involving germane load are due to the negated language used in items 7–9

Table 4: Factor Loadings and Reliability (Lecture 2)

Factor/Item	Factor Loading	Std Err	p-Value
IL - Intrinsic Load ($\alpha = 0.86$)			
Q1	0.77	0.050	< 0.001
Q2	0.95	0.037	< 0.001
Q3	0.75	0.047	< 0.001
EL - Extraneous Load ($\alpha = 0.85$)			
Q4	0.90	0.033	< 0.001
Q5	0.81	0.043	< 0.001
Q6	0.72	0.051	< 0.001
GL - Germane Load ($\alpha = 0.93$)			
Q7	0.85	0.033	< 0.001
Q8	0.91	0.026	< 0.001
Q9	0.89	0.032	< 0.001
Q10	0.83	0.035	< 0.001
Residual Covariance			
Q7 with Q9	-0.08*	0.156	0.620
Q9 with Q10	0.13*	0.143	0.361

*denotes a correlation rather than a loading

Table 5: Factor Correlations (Lecture 2), all significant at $p \leq 0.01$

	IL	EL	GL
IL	1.0	0.403	-0.272
EL		1.0	-0.739
GL			1.0

as compared to those on the rest of the survey (as previously discussed). With respect to the relationship between IL/GL, there was only low correlation ($r = -0.272$) accounting for just 7.4% of the co-variance between the two factors.

4.2 Lecture 2

Analysis of the data collected during a subsequent class lecture (N=117) also demonstrated excellent model fit ($\chi^2(30) = 39.7, p = 0.11; RMSEA = 0.053, pclose = 0.43; CFI = 0.99; TLI = 0.98$). As shown in Table 4, the standardized item loadings for this set of data followed a very similar pattern to that found with Lecture 1 responses. All items strongly loaded on their respective factors. Cronbach's alpha values demonstrated good internal consistency within the factors, and the overall reliability for the scale here was $\alpha = 0.87$. Again, none of the modification indices suggested further substantive additions to the model. Lastly, the two residual/error paths in the model (between items q7/9 and q9/10) were not significantly correlated here. Taken in combination with the low correlations between these items observed with the Lecture 1 data, they could likely be removed from the factor model entirely without negatively impacting overall model fit or its interpretation significantly. (Removing the two residual/error paths was originally suggested by [20] after analysis of Study 3; we have kept the co-variance pairs in an effort to be faithful to the original study.) Simplifying the model in this way and verifying it with additional data is one avenue of our future work.

The factors correlated (Table 5) in a nearly identical pattern to what was observed with the Lecture 1 data. Again IL and EL had a weak, positive correlation and EL and GL showed a moderate, negative correlation.

Overall, we conclude that the CS Cognitive Load Component Survey as adapted for introductory computer science validly replicates the original findings in statistics and can reliably reproduce them across multiple lectures in a CS1

class. In the next section we discuss the implications of measuring cognitive load components within a CS1 class.

5. DISCUSSION

The contribution of this paper is the creation of the instrument, adapted from an existing instrument for measuring cognitive load. We use the instrument to explore hypotheses about these two lectures, to demonstrate how the instrument might be used in experiments. A thorough exploration of the differences between the two lectures would require a deeper and broader analysis than just considering cognitive load.

To analyze the perceived cognitive load levels during each of the two lectures we needed to determine metrics for each of the factors. These metrics are computed as the average rating given all of the items within a factor and thus falls in the range 0–10. Table 6 presents the mean and standard deviation (σ) for each survey item and survey factor. Metrics from Lecture 1 are shown on the left-hand side of the table, while corresponding metrics from Lecture 2 are on the right.

We hypothesized that the lecture on sound processing (Lecture 2) should have posed a lower intrinsic load than the lecture on processing lists to produce HTML (Lecture 1). Recall that the IL component measures the innate connectedness or inherent difficulty of a topic (for a novice). The introduction of sound processing requires only the knowledge of sampling sounds to generate digital representations, while processing lists to create HTML code requires knowledge of strings, lists in Python, and valid HTML code. Analytically speaking, Lecture 1’s content requires holding multiple concepts in working memory at one time; more so than Lecture 2’s content. The data collected in this study supports this hypothesis, and IL during Lecture 1 was statistically significantly greater than during Lecture 2 ($t(271) = 4.43, p < 0.001$).

With respect to extraneous load, we also believed that Lecture 2 would pose lower demands on mental resources, and this was confirmed with statistical significance ($t(271) = 3.08, p = 0.002$). EL consists of instructional material that serves to distract the student from learning. Split attention between two elements, using only text explanations, and unnecessary redundant material all contribute to a higher EL score. In our case, we hypothesized that Lecture 2 would have a lower EL score because it makes use of dual input channels—both audio as well as visual inputs [13]. A known multimedia learning principle to reduce extraneous cognitive load is to use both pictures and sound rather than text only to explain a concept [13]. Even though students were most likely more familiar with HTML and web pages than with sound manipulation, the sound processing lecture produced the lower EL average. The sheer number of items being manipulated to produce the web page with Python code (Python editor window, HTML code editor window, browser window) forces split attention for the student. Even though with sound processing the editor window and sound visualization are both on screen, the student does not attempt to interpret or “read” the visualization; it is merely a placeholder and representation for the sound which is heard.

The final cognitive load factor is that of germane load, in which Lecture 1 had a statistically significantly lower score ($t(271) = -3.91, p < 0.001$). Any reduction in GL would indicate that students are utilizing too much working memory for IL and EL, thus reducing the amount of learning. According to cognitive load theory, students may have learned less from Lecture 1 than they did from Lecture 2. This could

Table 6: Average Factor Scores by Lecture

Q	Lecture 1 (N=156)			Lecture 2 (N=117)			
	Avg	σ	Factor Avg	Avg	σ	Factor Avg	
1	6.32	2.28	IL=6.33 $\sigma = 2.07$	5.18	2.38	IL=5.19 $\sigma = 2.16$	
2	6.35	2.36		5.21	2.49		
3	6.32	2.45		5.18	2.45		
4	3.67	2.45	EL=3.59 $\sigma = 2.11$	2.74	2.18	EL=2.81 $\sigma = 1.98$	
5	3.47	2.45		2.62	2.18		
6	3.62	2.60		3.07	2.41		
7	5.63	2.40	GL=5.54 $\sigma = 2.20$	6.63	2.07	GL=6.53 $\sigma = 1.87$	
8	5.58	2.40		6.38	2.05		
9	5.51	2.51		6.58	2.00		
10	5.47	2.43		6.52	2.14		

be determined through student performance on questions addressing the concepts covered during the two lectures. Because surveys were collected anonymously, correlation with test question performance was impossible here—and beyond our focus on the instrument itself. We discuss how correlation with performance could be done in the next section.

5.1 Future Work

Now that an instrument to measure cognitive load dimensions has been constructed, it can be used within other experimental situations.

The purpose of the CS Cognitive Load Component Survey (CS CLCS) is to compare pedagogical interventions. We can use it to evaluate multiple sets of instructional materials developed along with a hypothesis that one set of materials would be higher (or lower) in either EL (the most common) or IL. Participants could then be randomly assigned to a specific treatment. Upon completion of the learning activity, the participant could be asked to complete the CS CLCS. Metrics for each cognitive load factor can be calculated by averaging all the responses to the questions contributing to that factor. For example, calculate the average to all the responses for questions 1, 2, and 3 to calculate the IL.

The CS CLCS can also be combined with learner performance indicators such as retention or transfer questions. Here it would be important to have participants complete the CS CLCS immediately following knowledge acquisition or learning so as to measure the components of the training material specifically. Once the performance data and survey data have been collected, it should be analyzed to determine if the survey results correlate with the post-test results. Questions to be asked include:

- Does a higher GL score indicate better learning?
- Does a higher pre-test score affect the IL score?
- Does a higher EL score indicate lower performance?

Having the ability to correlate the cognitive load components and learning results gives the researcher the ability to test for and quantify specific elements of the instructional material for iterative design improvement.

The authors plan to use the CS CLCS to determine the cognitive load components for understanding code segment explanations via text or auditory explanations. It is currently unknown what the most effective modality for code segment explanations is for those just learning to program. We will use the CS CLCS along with performance indicators to explore how modality affects learning. Our process can be seen in Figure 3. CLT and multi-media principles suggest that auditory only explanations should yield a lower overall

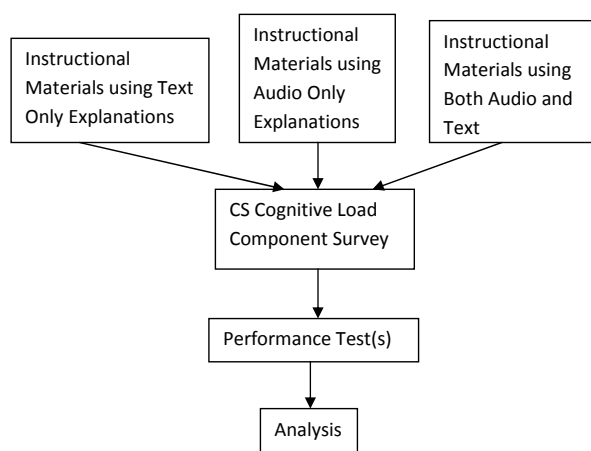


Figure 3: Using CS Cognitive Load Component Survey to Determine Modality Learning Effects

EL due to the use of dual channels [13], however this result remains to be proven empirically.

5.2 Limitations

While we have adapted and initially verified the use of the CS CLCS here, there are some limitations and opportunities for further survey refinement. We acknowledge that this study used data collected in two sections of a single introductory CS class with lectures given by the same instructor. These lectures were chosen in part because of opportunity (we wanted to ensure the same lecturer and identical lecture content) and because of the topics. Both lectures were the introduction of new topics. Wider data collection with other courses and instructors was done by [21], but the study has not yet been replicated in computer science.

Elements of our survey construction warrant further investigation. In particular, more experimentation should be conducted to explore different wordings of the questions in the survey. While our method here randomized question presentation to control for ordering-effects, item phrasing could be confounding the factor groupings we observed. It is possible that at least some of the inter-item co-variance was due to common stems across multiple items. For example, “The activity really enhanced my understanding of ...,” used in questions 7–9 which all contribute to the germane load factor. Further, in each set of factor questions, all the questions are positively (or negatively) worded. Such concerns about inadvertent bias due to wording are common in survey design; however, in this initial study we opted to use language that matched the original phrasing from Leppink et al [20] as closely as possible. In future follow-on studies, we plan to examine the impact of randomizing positively and negatively worded items within each subscale along with minimizing shared prompt substrings to the extent possible. Further, a qualitative study using a think-aloud protocol during survey completion would uncover how learners *interpret* these items which could aid in further item refinement.

6. CONCLUSION

The contribution of this paper is the adaptation and collection of initial validity evidence for an instrument to measure cognitive load components within introductory CS class

lectures. The high Cronbach’s alpha values, the high fit indices (i.e., the CFI and TLI), and the low RMSEA in the data for both lectures support the three-factor cognitive load structure of CLT and provide validity evidence for the CS Cognitive Load Component Survey in measuring cognitive load components in a CS1 class. We describe how the instrument could be used, along with student performance indicators, to investigate specific pedagogical interventions.

We know that computer science is difficult to learn. For example, the Learning Edge Momentum theory [31] suggests that the deep interconnections in CS topics leads to bimodality in grades. Deep interconnections are measured by IL. An instrument to measure cognitive load helps us to ask research questions about the amount of cognitive load and how we can manage the cognitive load through careful instructional design. We hope that CS CLCS (with further development) and its successors give computing education researchers a new tool for developing theory and better interventions to address the unique learning challenges of computer science.

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